Syed Moshiur Murshed

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Professional Summary

Senior Software Engineer and Technical Lead with 15+ years of experience building, evolving and operating commercial software products. Spent 14 years designing, extending and ultimately taking technical ownership of Redbex and MappRover, large-scale C#/.NET platforms used in civil infrastructure and engineering projects.

Strong background in C#/.NET, distributed systems, APIs, and data-intensive applications, with a focus on maintainable architecture, software quality, and long-term system evolution. Experienced in combining hands-on development with technical leadership, mentoring engineers, and improving engineering practices. Experience designing and maintaining REST and SOAP-based integration services for enterprise systems.

Comfortable working across backend services, APIs, data modelling, DevOps, and cloud-based systems (Azure).

Open to Senior, Staff, and Principal engineering roles combining hands-on development with technical leadership.

Key Strength: Bridging deep hands-on engineering, domain modelling, and software architecture — with the technical leadership to raise quality and system design across a team.

Signature Products

Redbex — Sensor Data Integration and Analysis Platform

Pöyry's commercial B2B platform for sensor data management across civil and energy engineering — built on OGC Sensor Web Enablement (SWE) standards, deployed as licensed self-hosted installations and as a Pöyry-managed hosted service. Used across civil infrastructure and energy engineering projects handling large time-series sensor datasets.

- Led the evolution of a metadata-driven engineering platform where new monitoring domains and measurement types could be introduced through configuration rather than software releases.
- Designed and owned the dynamic hierarchical view and filter system — the architectural core of the platform. Views are modelled as composable predicate trees with unlimited depth and dynamic membership evaluation: features join or leave views automatically as data or time changes, with no direct feature-to-view associations. The same mechanism drives reporting, GIS visualisation, analytics, exports, and access control across all deployments
- Designed and built the mapping module — a layered geospatial rendering engine supporting multiple configurable layer types, on-the-fly spatial reference system transformation, and persistable map view templates. Dual-mode: renders interactively in the Redbex UI and exposes a map rendering service producing embeddable images for external systems
- Architected and led a major refactoring of the data import framework — a two-stage pipeline (source extraction → normalised intermediate format → persistence) with a fully configurable generic import type, allowing clients to map any source column to any Redbex field without custom code per deployment
- Built and owned core platform modules — messaging, approval workflows, notifications and alerting, data quality and validation, and reporting and export — before progressively taking ownership of further modules including the charting engine, job execution engine, and CAD export pipeline, ultimately assuming technical ownership of the entire Redbex product
- Led SQL performance engineering: query tuning, index strategy, and data consistency improvements across large time-series sensor observation datasets and hierarchical domain models, significantly improving response times for reporting and GIS queries
- Designed and maintained SOAP-based integration services and APIs consumed by external clients, MappRover, and third-party integrations — establishing contract stability as platform-level concerns.
- Defined backend architecture patterns adopted across engineering teams, reducing integration friction and establishing a consistent foundation for domain-specific add-on development

MappRover — Offline-First Geological Mapping Application

iOS application (Xamarin) replacing pen-and-paper geological documentation during tunnel construction — geo-located mapping along a tunnel axis, fully operational without network connectivity, synced to the Redbex platform

- Designed the offline sync protocol with deterministic conflict handling — a hard requirement given the zero-connectivity environment of active tunnel construction sites, where engineers map geological characteristics directly inside the bore
- Owned the full integration layer between the MappRover iOS client and the Redbex Application Server, including the data synchronisation procedure reconciling offline edits with server-side state modified via the Redbex Smart Client

GUSS — Construction Industry Cost Estimation Platform

Domain-rich B2B application for the Austrian and German construction industry, specialising in cost estimation and service specification management

- Worked as lead developer on feature development and platform enhancements across a technically mature product with clear separation of concerns
- Contributed across a complex domain model covering construction cost structures and specification management, gaining depth in Austrian/German construction industry workflows

Technical Skills

- **Core Engineering:** C#, .NET Framework, .NET Core, ASP.NET Web API, WCF, REST & SOAP web services, backend services, distributed systems, Entity Framework (EF)
- **Cloud & DevOps:** Azure, Docker, CI/CD, Azure DevOps, Git
- **Data:** SQL Server, PostgreSQL, data modelling, query optimisation, performance tuning, indexing strategies
- **Architecture & Design:** System design, API design, domain modelling, Domain-Driven Design (DDD), CQRS, software architecture, maintainable systems, integration architecture
- **Desktop & Mobile:** WPF, WinForms, Xamarin (iOS/iPad), MVVM
- **Other:** GIS integration, system integration, technical mentoring

Professional Experience

Head of Software Development (Technical IC Track)

AFRY Austria GmbH (Former Pöyry Austria GmbH), Salzburg, Austria ***01/2020 – 07/2025***

- Combined technical leadership with hands-on ownership of ***Redbex*** and ***MappRover***, leading architecture, backend services, APIs, and long-term product evolution across multiple production deployments.
- Led and mentored a team of developers, supporting growth in system design, code quality, and engineering practices; conducted code reviews and guided architectural decisions
- Drove platform modernisation by refactoring core subsystems, improving maintainability and scalability without disrupting live client systems
- Partnered closely with product owners and stakeholders to translate complex requirements into reliable, maintainable solutions, including hiring and onboarding, helping scale the team and establish consistent engineering standards
- Maintained hands-on ownership of critical backend services, integrations, and platform architecture while balancing technical leadership responsibilities.
- Served as Lead Developer and technical owner in a cross-team setup with AFRY Sweden, designing cloud-native APIs and distributed business services for ***(IndustryX, MaterialX)*** platforms using ASP.NET Core, DDD, CQRS, PostgreSQL, Docker and GitLab CI/CD.
- Led pilot projects including automation and ML-based processing initiatives, coordinating between teams, clients, and external partners.

Lead Software Engineer

Pöyry Austria GmbH, Salzburg, Austria ***06/2018 – 01/2020***

- Led design and implementation of major subsystems in ***Redbex*** and ***MappRover***, taking ownership from architecture through to production delivery

- Introduced and improved CI/CD pipelines (***Azure DevOps***), increasing delivery reliability and team consistency
- Mentored developers and promoted adoption of better engineering practices and modern design approaches
- Contributed actively to backend services, APIs, and key system architecture decisions

Software Engineer

Pöyry Austria GmbH, Salzburg, Austria — ***04/2011 – 06/2018***

- Core contributor to ***Redbex*** platform across full development lifecycle — analysis, architecture, implementation, and testing
- Designed and built foundational modules: messaging system, approval workflow engine, and GIS-based map components
- Established and maintained CI build infrastructure for stable, repeatable delivery cycles

Earlier Career

Master Thesis · Carl Zeiss AG, Oberkochen, Germany — ***04/2010 – 11/2010***

GPU-Accelerated Lens Polishing Optimisation for Industrial Automation

Designed and implemented OpenCL-based algorithms integrated into C#/.NET industrial systems, reducing computation time by 65% and improving surface quality metrics in production.

Student Programmer · Poolarserver Schwägerl & Friedle GbR, Stuttgart, Germany — ***08/2009– 04/2010***

Developed and maintained components of an ASP.NET (MVC) web platform with clean separation of concerns, building performant client–server interactions using jQuery, AJAX, JSON, and LINQ-to-SQL with SQL Server. Stack: ASP.NET, jQuery, AJAX, JSON, LINQ-to-SQL, C#.

Software Engineer · Adaptive Enterprise Limited, Dhaka, Bangladesh — ***03/2006 – 09/2008***

Full-stack .NET engineer on ***CardAccess*** legacy migration (Delphi → .NET), building WPF/MVVM facility monitoring systems, a script server with C# code compilation, and an AJAX configuration portal. Cross-continental team with US stakeholders.

Freelance Programmer · Dhaka, Bangladesh — ***04/2004 – 02/2006***

Built multiple web applications for telecoms clients including a document management system, task logger, and registration management platform. Stack: ASP.NET, C#, SQL Server.

Education

M.Sc. in Software Technology

Stuttgart University of Applied Sciences, Stuttgart, Germany — ***03/2009 – 11/2010***

B.Sc. in Computer Science

American International University, Dhaka, Bangladesh — ***10/2000 – 08/2006***

Languages

- English: Fluent
- German: B2
- Bengali: Native
- Hindi: Basic

References

Johannes Dölzlmüller (former MD AFRY Austria GmbH),

contact details available on request

Aamir Faried (former Group Manager and line manager at AFRY),

contact details available on request